

Quantum Mechanics

* Time dependent Schrodinger's Equation.

One-D wave equation for a particle:

$$\frac{d^2 y}{dx^2} = \frac{1}{v^2} \frac{d^2 y}{dt^2} \rightarrow (1)$$

For a microscopic particle;

→ Equation (1) can be modified as,

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \rightarrow (2)$$

The solution for eqⁿ (2) can be written as

$$\psi(x, t) = A e^{i(kx - \omega t)} \rightarrow (3)$$

k = propagation vector

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{h/p} = \frac{2p\pi}{h} = \frac{p}{\hbar} \left\{ \hbar = \frac{h}{2\pi} \right\} \rightarrow (4)$$

$$\omega = 2\pi \nu \xrightarrow{\text{nu}} = \frac{2\pi E}{h} = \frac{E}{\hbar} \rightarrow (5)$$

Substitute eqⁿ (4) & (5) in eqⁿ (3), we get

$$\psi(x, t) = A e^{\frac{i}{\hbar} (Px - Et)} \rightarrow (6)$$

$$\psi(x, t) = A \exp \left[\frac{i}{\hbar} (Px - Et) \right]$$

The total energy associated with a particle is given as

$$TE = KE + PE \rightarrow (7)$$

$$\rightarrow E = \frac{1}{2} mv^2 + V$$

$$\rightarrow E = \frac{p^2}{2m} + V \rightarrow (8)$$

Differentiate eqⁿ (6) wrt x ;

$$\frac{\partial \psi}{\partial x} = A e^{\frac{i}{\hbar} (px - Et)} \cdot \left(\frac{i p}{\hbar} \right)$$

$$\frac{\partial \psi}{\partial x} = \frac{i p}{\hbar} \psi \rightarrow (9)$$

Differentiating eqⁿ (9) again wrt x

$$\rightarrow \frac{\partial^2 \psi}{\partial x^2} = A e^{\frac{i}{\hbar} (px - Et)} \cdot \left(\frac{i p}{\hbar} \right)^2$$

$$\frac{\partial^2 \psi}{\partial x^2} = - \frac{p^2}{\hbar^2} \psi$$

~~Multiple~~ Rearrange the term

$$- \hbar^2 \frac{\partial^2 \psi}{\partial x^2} = p^2 \psi$$

Divide both sides by $2m$

$$\frac{-\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = \frac{p^2}{2m} \psi \rightarrow (10)$$

Now differentiate eqⁿ (6) wrt time function

$$\begin{aligned} \rightarrow \frac{\partial \psi}{\partial t} &= A e^{\frac{i}{\hbar} (px - Et)} \cdot \left(\frac{-iE}{\hbar} \right) \\ &= \frac{-iE}{\hbar} \psi \end{aligned}$$

$$\frac{\partial \psi}{\partial t} = \frac{-i^2 E}{i\hbar} \psi = \frac{E}{i\hbar} \psi$$

$$\rightarrow E \psi = i\hbar \frac{\partial \psi}{\partial t} \rightarrow (11)$$

Eqⁿ (8) can be written as

$$E \psi = \frac{p^2}{2m} \psi + V \psi \rightarrow (12)$$

as in QM P.E & V are differential operators

Substitute eqⁿ (10) & (11) in eqⁿ (12)

$$\rightarrow \frac{-\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V \psi = \frac{i\hbar \partial \psi}{\partial t} \rightarrow (13)$$

Time dependent form of Schrodinger's eqⁿ / TDSE

* Time Independent Schrodinger wave Eqⁿ

(TISE)

TOSE:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi = i\hbar \frac{\partial \Psi}{\partial t} \quad (13)$$

Here, $\Psi = \Psi(x, t)$

Use the method of separation of variables

$$\Psi(x, t) = \Psi(x) \phi(t) \rightarrow (14)$$

written as multiplication of two separate functions

Substitute (14) in (13)

$$\rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x) \phi(t)}{\partial x^2} + V(\Psi(x) \phi(t)) = i\hbar \frac{\partial \Psi(x) \phi(t)}{\partial t} \quad (15)$$

$$\rightarrow -\frac{\hbar^2}{2m} \phi(t) \frac{\partial^2 \Psi(x)}{\partial x^2} + V\Psi(x)\phi(t) = i\hbar \Psi(x) \frac{\partial \phi(t)}{\partial t} \rightarrow (16)$$

Divide (16) by (14)

$$\rightarrow \underbrace{-\frac{\hbar^2}{2m} \frac{1}{\Psi(x)} \frac{\partial^2 \Psi(x)}{\partial x^2} + V}_{\text{function of } x} = \underbrace{i\hbar \frac{1}{\phi(t)} \frac{\partial \phi(t)}{\partial t}}_{\text{function of } t = E \text{ operator}} \rightarrow (17)$$

LHS = RHS = constant

$$-\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{d^2(\psi(x))}{dx^2} + V = E \rightarrow (18)$$

Multiply eqⁿ (18) by $\psi(x)$

$$\rightarrow -\frac{\hbar^2}{2m} \frac{d^2(\psi(x))}{dx^2} + V\psi(x) = E\psi(x) \rightarrow (19)$$

* Writing TISE in 3.D. form

$$\rightarrow -\frac{\hbar^2}{2m} \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right] \psi(x, y, z) + V\psi(x, y, z) = E\psi(x, y, z) \rightarrow (20)$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(x, y, z) + V\psi(x, y, z) = E\psi(x, y, z) \rightarrow (21)$$

$$\vec{\nabla} = \frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$$

$$\nabla^2 = \vec{\nabla} \cdot \vec{\nabla} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\rightarrow \underbrace{\left[-\frac{\hbar^2}{2m} + V \right]}_{\rightarrow H} \psi(x, y, z) = E\psi(x, y, z)$$

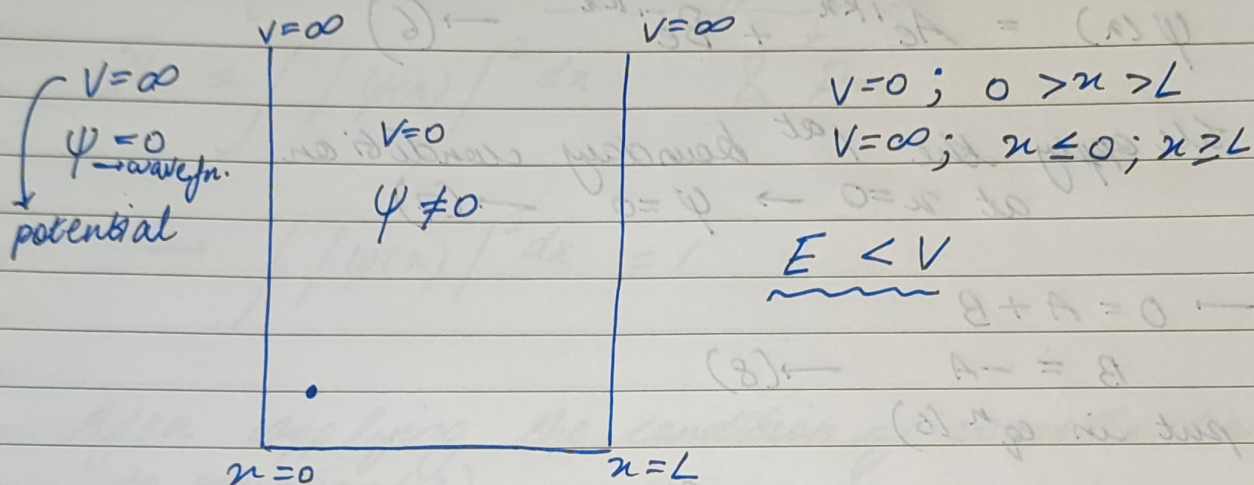
$$\rightarrow H\psi = E\psi \rightarrow (22)$$

where H = Hamiltonian operator

$$H = \left[-\frac{\hbar^2}{2m} \nabla^2 + V \right]$$

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* Infinite 1-D Potential well/Particle in a box



We start with TISE

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V\psi = E\psi \rightarrow (1)$$

$$\rightarrow \frac{d^2 \psi}{dx^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0 \rightarrow (2)$$

Now, as the particle is trapped in the box, $V = 0$
 $\rightarrow$ eqⁿ (2) can be written as;

$$\frac{d^2 \psi}{dx^2} + \frac{2m}{\hbar^2} E\psi = 0 \rightarrow (3)$$

$$\text{Assume } \frac{2mE}{\hbar^2} = k^2 \rightarrow (4)$$

eqⁿ (4) in (3)

$$\rightarrow \frac{d^2 \psi}{dx^2} + k^2 \psi = 0 \rightarrow (5)$$

The solution of the eqⁿ can be given as

$$\psi(x) = Ae^{ikx} + Be^{-ikx} \rightarrow (6)$$

We apply the 1st boundary condition at $x=0 \rightarrow \psi=0 \rightarrow (7)$

$$\rightarrow 0 = A + B$$

$$B = -A \rightarrow (8)$$

put in eqⁿ (6)

$$\begin{aligned} \rightarrow \psi(x) &= Ae^{ikx} - Ae^{-ikx} \\ &= A(e^{ikx} - e^{-ikx}) \end{aligned}$$

$$= 2iA \left[\frac{e^{ikx} - e^{-ikx}}{2i} \right] \rightarrow (9)$$

$$\psi(x) = 2iA \sin kx \rightarrow (10)$$

Applying 2nd boundary condition to (10) at $x=L \rightarrow \psi=0$

Eqⁿ (10) can be modified as;

$$\rightarrow 0 = 2iA \sin kL$$

$$\rightarrow \sin kL = 0 \rightarrow kL = \sin^{-1}(0) \quad \text{we don't accept value 0}$$

$$\rightarrow kL = n\pi$$

$$\rightarrow k = \frac{n\pi}{L} \rightarrow (11)$$

put (11) in (10)

$$\rightarrow \psi(x) = 2iA \sin \left(\frac{n\pi x}{L} \right) \rightarrow (12)$$

Now in order to get the value of constant $2iA$, we apply the condition of normalisation

$$P = \int |\psi(x)|^2 dx \quad \& \quad P=1$$

$$\rightarrow \int |\psi(x)|^2 dx = 1$$

After applying the condition of normalisation, to equation (12)

$$\text{we get } 2iA = \sqrt{\frac{2}{L}} \rightarrow (13)$$

when (13) is substituted in (12), we get

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \rightarrow (14)$$

From eqⁿ (4) we know,

$$\frac{2mE}{\hbar^2} = k^2 \rightarrow E = \frac{\hbar^2 k^2}{2m}$$

$$\text{As } \hbar = \frac{h}{2\pi} \quad \& \quad k = \frac{n\pi}{L}$$

$$\rightarrow E_n = \frac{n^2 h^2}{8mL^2}$$

n : no. of energy state

h : planck's const

m : mass of particle trapped in box

L : width of the box

Energy is quantized

Energy levels are discrete

constant

$\rightarrow E_n$ depends on n^2